Computer Vision System & Processing Pipeline

To maintain high responsiveness at speed, our vision system trades unnecessary resolution for processing frequency:

[Raw Camera Feed: QQVGA @ 70 FPS]

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[YUV422 Color Space Capture]

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[Discard Y Channel (Luminance Isolation)]

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[Apply Threshold Masks on U & V Channels]

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[NumPy Vectorized Pixel Mean (Centroid X, Y)]

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[Calculate Relative Angle & Distance Scale]

- **High-Speed Framing:** The camera is configured to output **QQVGA resolution at 70 FPS**.
- **YUV422 Processing Optimization:** Instead of converting frames to BGR or HSV, raw **YUV422** frames are captured directly. The luminance channel (Y) is discarded to build lighting-invariant color masks using only the chrominance channels (U and V).
- **Vectorized Centroid Calculation:** NumPy computes the mean X and Y coordinates across all masked pixels, returning the exact centroid of red and green pillars in milliseconds without heavy contour-fitting operations.